Problem 2(ii): From QED to QCD

Deriving the $q\,q' \to q\,q'$ Matrix Element from $e^-\mu^- \to e^-\mu^-$

PHYS 7502 — Particles and Nuclei 2

1 Question and Deliverables

Problem 2(ii)

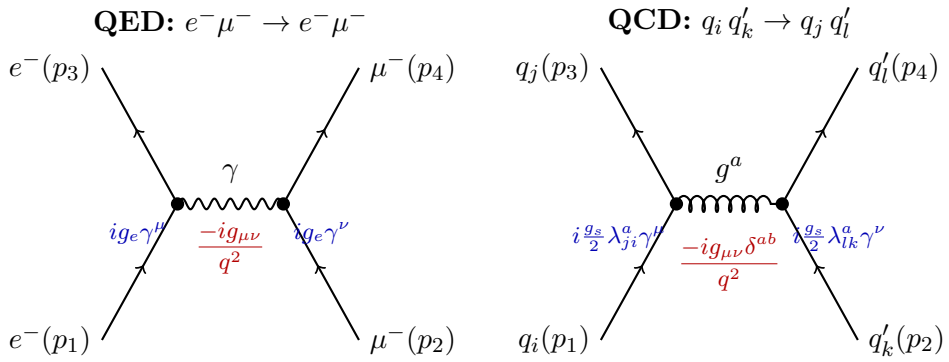
Convert the Mandelstam variable amplitude to the equivalent one for $qq \to qq$ parton–parton scattering via gluon exchange. Presume different quark flavors to avoid the u-channel. You may either sum/average over colors, or choose specific color assignments.

Deliverables

- Derive the QED matrix element $\mathcal{M}$ for $e^-\mu^- \to e^-\mu^-$ from Feynman rules.
- Derive the QCD matrix element $\mathcal{M}$ for $q\,q' \to q\,q'$ from Feynman rules.
- Show line-by-line how every QCD piece maps onto its QED analog.
- Square, spin-sum, color-sum/average, and express in Mandelstam variables.

2 The Two Diagrams Side by Side

Both processes have exactly one lowest-order diagram: a t-channel exchange of a massless spin-1 boson between two distinct fermion lines.



Why the Same Topology?

Both diagrams are t-channel because the two fermion lines have different identity: $e^- \neq \mu^-$ in QED, $q \neq q'$ (different flavor) in QCD. There is no way for the electron line to "become" a muon line (or vice versa), and likewise a u-quark cannot become a d-quark at a gluon vertex. So there is no u-channel diagram (which would require swapping two identical outgoing particles) and no s-channel diagram (which would require annihilation of the two different fermions into the mediator).

3 Step-by-Step: Writing $\mathcal{M}$ from the Feynman Rules

We now build each amplitude ingredient-by-ingredient, following the fermion lines *backward* (against the arrow), as prescribed by Griffiths' Rule 7 (2nd ed.).

3.1 The QED Amplitude

Building $\mathcal{M}_{\text{QED}}$ Piece by Piece (Griffiths 2nd ed.

Fermion line 1 (electron): Follow the arrow backward from the outgoing $e^-(p_3)$ to the incoming $e^-(p_1)$.

- Start with adjoint spinor for outgoing electron: $\bar{u}(p_3)$
- Hit vertex $\rightarrow$ write vertex factor: $ig_e\gamma^\mu$
- End with spinor for incoming electron: $u(p_1)$

Result: $\bar{u}(p_3)\,(ig_e\gamma^\mu)\,u(p_1)$

Fermion line 2 (muon): Same procedure.

- Adjoint spinor for outgoing muon: $\bar{u}(p_4)$
- Vertex factor: $ig_e\gamma^\nu$
- Spinor for incoming muon: $u(p_2)$

Result: $\bar{u}(p_4)\,(ig_e\gamma^\nu)\,u(p_2)$

Photon propagator connecting the two vertices (indices μ, ν contract):

$$\frac{-ig_{\mu\nu}}{q^2}$$

where $q = p_1 - p_3$, so $q^2 = (p_1 - p_3)^2 = t$ (the Mandelstam variable).

Assemble (multiply everything, cancel the overall δ-function, set $= -i\mathcal{M}$):

$$-i\mathcal{M}_{\text{QED}} = \left[\bar{u}(3)\,(ig_e\gamma^\mu)\,u(1)\right]\,\frac{-ig_{\mu\nu}}{t}\,\left[\bar{u}(4)\,(ig_e\gamma^\nu)\,u(2)\right]$$

$$= \frac{-i\,g_e^2}{t}\,\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right] \tag{1}$$

(The i's combine: $i\cdot(-i)\cdot i = -i$, and the $g_{\mu\nu}$ lowers the index $\nu \rightarrow \mu$.)

Dividing both sides by $-i$:

$$\boxed{\mathcal{M}_{\text{QED}} = \frac{g_e^2}{t}\,\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right]} \tag{2}$$

(Griffiths 2nd ed., eq. 7.106, with $q^2 = (p_1 - p_3)^2 = t$ and sign conventions absorbed.)

3.2 The QCD Amplitude

Building $\mathcal{M}_{\text{QCD}}$ Piece by Piece (Griffiths 2nd ed.

Fermion line 1 (quark of flavor q, color $i \to j$): Follow backward.

- Adjoint spinor for outgoing quark: $\bar{u}(p_3)$
- Hit vertex $\to$ QCD vertex factor: $i\dfrac{g_s}{2}\lambda^a_{ji}\gamma^\mu$

  (The λ^a_{ji} is just a *number*—the (j,i) matrix element of the a-th Gell-Mann matrix. It tells you how the quark's color rotates: incoming color i, outgoing color j, gluon color a.)
- Spinor for incoming quark: $u(p_1)$

Result: $\bar{u}(p_3)\left(i\frac{g_s}{2}\lambda^a_{ji}\gamma^\mu\right)u(p_1) = i\frac{g_s}{2}\lambda^a_{ji}\left[\bar{u}(3)\gamma^\mu u(1)\right]$

Fermion line 2 (quark of flavor q', color $k \to l$): Same.
Result: $i\frac{g_s}{2}\lambda^b_{lk}\left[\bar{u}(4)\gamma^\nu u(2)\right]$

Gluon propagator (color indices a, b; Lorentz indices μ, ν):

$$\frac{-ig_{\mu\nu}\,\delta^{ab}}{q^2}$$

The δ^{ab} is **new compared to QED**. It enforces that the gluon emitted at vertex 1 (color a) is the same gluon absorbed at vertex 2 (color b). The photon propagator has no such factor because photons don't carry charge.

Assemble:

$$-i\mathcal{M}_{\text{QCD}} = \left[i\frac{g_s}{2}\lambda^a_{ji}\,\bar{u}(3)\gamma^\mu u(1)\right]\frac{-ig_{\mu\nu}\delta^{ab}}{t}\left[i\frac{g_s}{2}\lambda^b_{lk}\,\bar{u}(4)\gamma^\nu u(2)\right] \tag{3}$$

Let's track each factor:

- i**'s**: $i\cdot(-i)\cdot i = -i$. ✓ (same as QED)
- **Couplings**: $\frac{g_s}{2}\cdot\frac{g_s}{2} = \frac{g_s^2}{4}$
- **Color**: δ^{ab} collapses $\lambda^a\lambda^b \to \sum_a \lambda^a_{ji}\lambda^a_{lk}$
- **Lorentz**: $g_{\mu\nu}$ lowers $\nu \to \mu$, giving γ_μ on the second bracket
- **Spinors**: Identical structure to QED

$$\boxed{\mathcal{M}_{\text{QCD}} = \frac{g_s^2}{4t}\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right]\sum_{a=1}^{8}\lambda^a_{ji}\,\lambda^a_{lk}} \tag{4}$$

(Griffiths 2nd ed., eq. 8.56.)

4 Comparing QED and QCD: What Changed?

Line-by-Line Comparison

Ingredient	**QED** ($\mathcal{M}_{\text{QED}}$)	**QCD** ($\mathcal{M}_{\text{QCD}}$)
Overall coupling	g_e^2	$\frac{g_s^2}{4}$
Propagator denominator	$\frac{1}{t}$	$\frac{1}{t}$ (identical)
Fermion line 1	$\bar{u}(3)\gamma^\mu u(1)$	$\bar{u}(3)\gamma^\mu u(1)$ (identical)
Fermion line 2	$\bar{u}(4)\gamma_\mu u(2)$	$\bar{u}(4)\gamma_\mu u(2)$ (identical)
Color factor	**None** (photon is colorless)	$\sum_{a=1}^{8} \lambda^a_{ji}\,\lambda^a_{lk}$

The relationship is:

$$\mathcal{M}_{\text{QCD}} = \frac{g_s^2/4}{g_e^2} \times \sum_a \lambda^a_{ji}\lambda^a_{lk} \times \mathcal{M}_{\text{QED}} \tag{5}$$

The **entire Dirac structure is inherited**. Only the coupling and the color overlay change.

Why the Dirac Structure is Identical

The γ^μ matrices act in 4-dimensional **spinor space** (they shuffle the components of the Dirac spinor u). The λ^a matrices act in 3-dimensional **color space** (they shuffle the r/g/b components of the color vector c). These are completely independent vector spaces — a γ matrix never "sees" a λ matrix, and vice versa. So when we square and trace, the spinor traces and color sums factorize perfectly:

$$|\mathcal{M}_{\text{QCD}}|^2 = \underbrace{(\text{spinor traces})}_{\text{same as QED}} \times \underbrace{(\text{color sums})}_{\text{new in QCD}}$$

This factorization is the key insight that makes the whole problem tractable.

5 Step 1: Square the Amplitude and Sum over Spins

We now compute $\langle|\mathcal{M}|^2\rangle$ — averaged over initial spins *and* initial colors, summed over final spins *and* final colors.

5.1 The QED Spin Traces (Review)

Starting from $\mathcal{M}_{\text{QED}}$, squaring gives:

$$|\mathcal{M}_{\text{QED}}|^2 = \frac{g_e^4}{t^2}\left|\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right]\right|^2 \tag{6}$$

Apply Casimir's trick (Griffiths 2nd ed., eq. 7.123) to each fermion line — sum over spins, replace $\sum_s u\bar{u} \to \not{p} + mc$ (completeness relation):

$$\sum_{\text{spins}} |\mathcal{M}_{\text{QED}}|^2 = \frac{g_e^4}{t^2} \underbrace{\text{Tr}\big[\gamma^\mu(\not{p}_1 + m_1 c)\gamma^\nu(\not{p}_3 + m_3 c)\big]}_{\text{electron trace}} \underbrace{\text{Tr}\big[\gamma_\mu(\not{p}_2 + m_2 c)\gamma_\nu(\not{p}_4 + m_4 c)\big]}_{\text{muon trace}} \tag{7}$$

(Griffiths 2nd ed., eq. 7.126.)

How Casimir's Trick Works (Quick Reminder)

For a bilinear $[\bar{u}(a)\,\Gamma_1\,u(b)][\bar{u}(a)\,\Gamma_2\,u(b)]^*$, summing over spins of particles a and b gives:

$$\sum_{s_a, s_b} = \text{Tr}\big[\Gamma_1(\not{p}_b + m_b c)\bar{\Gamma}_2(\not{p}_a + m_a c)\big]$$

where $\bar{\Gamma} = \gamma^0\Gamma^\dagger\gamma^0$. For $\Gamma = \gamma^\mu$, we get $\bar{\Gamma} = \gamma^\mu$ (it's its own "conjugate"). The completeness relation $\sum_s u(p,s)\bar{u}(p,s) = \not{p} + mc$ converts the spin sum into a trace of γ-matrices — no spinors left!

In the massless limit ($m_1 = m_2 = m_3 = m_4 = 0$), we use the trace theorem (Griffiths Rule 13):

$$\text{Tr}[\gamma^\mu \not{a} \gamma^\nu \not{b}] = 4\big[a^\mu b^\nu + b^\mu a^\nu - g^{\mu\nu}(a \cdot b)\big]$$

Electron trace:

$$\text{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3] = 4\big[p_1^\mu p_3^\nu + p_3^\mu p_1^\nu - g^{\mu\nu}(p_1 \cdot p_3)\big] \tag{8}$$

Muon trace:

$$\text{Tr}[\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_4] = 4\big[p_{2\mu} p_{4\nu} + p_{4\mu} p_{2\nu} - g_{\mu\nu}(p_2 \cdot p_4)\big] \tag{9}$$

Contract the μ, ν indices between the two traces. This is a lengthy but mechanical calculation. The result is:

$$\begin{aligned} &\text{Tr}[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3]\ \text{Tr}[\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_4] \\ &\quad = 16\big[2(p_1 \cdot p_2)(p_3 \cdot p_4) + 2(p_1 \cdot p_4)(p_2 \cdot p_3)\big] \\ &\quad = 32\big[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3)\big] \end{aligned} \tag{10}$$

Where Does the 32 Come From?

Each trace produces a factor of 4 (from Rule 13). So $4 \times 4 = 16$ from the two traces. Then the contraction over μ, ν yields two surviving terms (the cross terms with $g^{\mu\nu} g_{\mu\nu} = 4$ cancel in a specific way), each with a factor of 2. So $16 \times 2 = 32$.

5.2 Convert to Mandelstam Variables

The Mandelstam variables for $1 + 2 \to 3 + 4$ are:

$$\begin{aligned} s &= (p_1 + p_2)^2 = 2p_1 \cdot p_2 \qquad \text{(massless)} \\ t &= (p_1 - p_3)^2 = -2p_1 \cdot p_3 \\ u &= (p_1 - p_4)^2 = -2p_1 \cdot p_4 \end{aligned} \tag{11}$$

with $s + t + u = 0$.

From these, the dot products in eq. (10) are:

$$p_1 \cdot p_2 = \frac{s}{2}, \quad p_3 \cdot p_4 = \frac{s}{2}, \quad p_1 \cdot p_4 = -\frac{u}{2}, \quad p_2 \cdot p_3 = -\frac{u}{2} \tag{12}$$

Substituting:

$$32\Big[\frac{s}{2} \cdot \frac{s}{2} + \big(-\frac{u}{2}\big)\big(-\frac{u}{2}\big)\Big] = 32 \cdot \frac{s^2 + u^2}{4} = 8(s^2 + u^2) \tag{13}$$

So the spin-summed QED result is:

$$\sum_{\text{spins}} |\mathcal{M}_{\text{QED}}|^2 = \frac{g_e^4}{t^2} \cdot 8(s^2 + u^2) \tag{14}$$

Spin-averaging (divide by $(2s_1 + 1)(2s_2 + 1) = 2 \times 2 = 4$):

$$\boxed{\langle |\mathcal{M}|^2 \rangle_{\text{QED}} = \frac{1}{4} \cdot \frac{8\, g_e^4 (s^2 + u^2)}{t^2} = \frac{2\, g_e^4 (s^2 + u^2)}{t^2}} \tag{15}$$

This is the result from part (i), now derived from scratch.

6 Step 2: The QCD Amplitude Squared

Now we do the same for QCD. Starting from:

$$\mathcal{M}_{\text{QCD}} = \frac{g_s^2}{4t} \left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right] \sum_a \lambda^a_{ji}\lambda^a_{lk}$$

Squaring:

$$|\mathcal{M}_{\text{QCD}}|^2 = \frac{g_s^4}{16\, t^2} \left|\left[\bar{u}(3)\gamma^\mu u(1)\right]\left[\bar{u}(4)\gamma_\mu u(2)\right]\right|^2 \times \left|\sum_a \lambda^a_{ji}\lambda^a_{lk}\right|^2 \tag{16}$$

Why the Color Factor Pulls Out as a Separate $|\cdots|^2$

The λ^a_{ji} factors are just *numbers* (complex matrix elements, though for Gell-Mann matrices they happen to be real). They multiply the spinor brackets as overall scalars. So when we take $|\mathcal{M}|^2$, they factor out cleanly:

$$\left|A \cdot B \cdot C_{\text{color}}\right|^2 = |A|^2 \cdot |B|^2 \cdot |C_{\text{color}}|^2$$

where A and B are the two spinor brackets. The color factor $|C_{\text{color}}|^2$ is then handled independently from the Dirac traces.

The Dirac part $|\,[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]\,|^2$ is **exactly the same** as in QED. Summing over spins via Casimir's trick gives the same trace result:

$$\sum_{\text{spins}} \left|[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]\right|^2 = 8(s^2 + u^2)/t^0 = 8(s^2 + u^2) \tag{17}$$

(The $1/t^2$ is already out front in eq. (16).)

So:

$$\sum_{\text{spins}} |\mathcal{M}_{\text{QCD}}|^2 = \frac{g_s^4}{16\, t^2} \cdot 8(s^2 + u^2) \times \left|\sum_a \lambda^a_{ji}\lambda^a_{lk}\right|^2 \tag{18}$$

Now we need to handle the color.

7 Step 3: The Color Factor (Sum/Average)

We want the **color-averaged** quantity: average over initial colors i, k (the colors of the two incoming quarks), sum over final colors j, l and the gluon index a.

7.1 Setting Up the Sum

Write out $|C_{\text{color}}|^2$ explicitly:

$$\left|\sum_a \lambda^a_{ji}\lambda^a_{lk}\right|^2 = \sum_{a,b} \lambda^a_{ji}\lambda^a_{lk}\,\left(\lambda^b_{ji}\right)^*\left(\lambda^b_{lk}\right)^* \tag{19}$$

Step 3a: Use Hermiticity of Gell-Mann Matrices

The Gell-Mann matrices satisfy $\lambda^{a\dagger} = \lambda^a$ (they are Hermitian). For any Hermitian matrix, $(\lambda^b_{ji})^* = \lambda^b_{ij}$. (Taking the complex conjugate of a matrix element is the same as transposing, and for Hermitian matrices transpose = original.)
So:

$$\left(\lambda^b_{ji}\right)^* = \lambda^b_{ij}, \qquad \left(\lambda^b_{lk}\right)^* = \lambda^b_{kl} \tag{20}$$

The color factor becomes:

$$\sum_{a,b} \lambda^a_{ji}\lambda^b_{ij}\;\lambda^a_{lk}\lambda^b_{kl} \tag{21}$$

Now sum over the color indices i, j (all values 1,2,3) and separately k, l:

Step 3b: Recognize the Matrix Traces

$$\sum_{i,j} \lambda^a_{ji}\lambda^b_{ij} = \sum_j \left(\sum_i \lambda^b_{ij}\lambda^a_{ji}\right) = \sum_j (\lambda^b\lambda^a)_{jj} = \text{Tr}(\lambda^b\lambda^a) \tag{22}$$

$$\sum_{k,l} \lambda^a_{lk}\lambda^b_{kl} = \sum_l \left(\sum_k \lambda^b_{kl}\lambda^a_{lk}\right) = \sum_l (\lambda^b\lambda^a)_{ll} = \text{Tr}(\lambda^b\lambda^a) \tag{23}$$

This is the crucial step: summing over both color indices of a product of two λ matrices is, by definition, a **trace**.

So the full color sum is:

$$\sum_{\text{colors}} |C_{\text{color}}|^2 = \sum_{a,b} \text{Tr}(\lambda^a\lambda^b)\;\text{Tr}(\lambda^a\lambda^b) = \sum_{a,b} \left[\text{Tr}(\lambda^a\lambda^b)\right]^2 \tag{24}$$

Step 3c: Apply the Trace Orthogonality Identity

The Gell-Mann matrices satisfy (Griffiths 2nd ed., eq. 8.52):

$$\text{Tr}(\lambda^a\lambda^b) = 2\,\delta^{ab} \tag{25}$$

This says the eight λ^a are "orthogonal" under the trace inner product, with normalization 2.

Quick check: $\lambda^3 = \text{diag}(1, -1, 0)$, so $(\lambda^3)^2 = \text{diag}(1, 1, 0)$, and $\text{Tr}(\lambda^3\lambda^3) = 1 + 1 + 0 = 2$. ✓

Substituting:

$$\sum_{a,b}\left[2\delta^{ab}\right]^2 = \sum_{a,b} 4\,\delta^{ab}\delta^{ab} = 4\sum_{a=1}^{8}(\delta^{aa})^2 = 4\sum_{a=1}^{8} 1 = 4\times 8 = 32 \tag{26}$$

7.2 Including the Averaging Factors

Step 3d: Divide by the Number of Initial States

We average over:

- **Spins**: $\frac{1}{(2\cdot\frac{1}{2}+1)^2} = \frac{1}{4}$ (two spin-$\frac{1}{2}$ particles)
- **Colors**: $\frac{1}{N_c\times N_c} = \frac{1}{3\times 3} = \frac{1}{9}$ (each quark has 3 possible colors)

We also need $\frac{1}{16}$ from the $(g_s/2)^4$ coupling, which is already out front as $g_s^4/16$.

Putting it all together:

$$\langle|\mathcal{M}|^2\rangle_{\text{QCD}} = \underbrace{\frac{1}{4}}_{\text{spin avg.}} \times \underbrace{\frac{1}{9}}_{\text{color avg.}} \times \frac{g_s^4}{16\,t^2} \times \underbrace{8(s^2+u^2)}_{\text{Dirac traces}} \times \underbrace{32}_{\text{color sum}}$$

$$= \frac{1}{4}\cdot\frac{1}{9}\cdot\frac{g_s^4}{16}\cdot\frac{8\cdot 32\,(s^2+u^2)}{t^2} \tag{27}$$

Compute the numerical prefactor step by step:

$$\frac{1}{4}\times\frac{1}{9}\times\frac{1}{16}\times 8\times 32 = \frac{8\times 32}{4\times 9\times 16} = \frac{256}{576} = \frac{4}{9} \tag{28}$$

Final Result

$$\boxed{\langle|\mathcal{M}|^2\rangle_{qq'\to qq'} = \frac{4}{9}\,g_s^4\,\frac{s^2+u^2}{t^2}} \tag{29}$$

Equivalently, with $\alpha_s = g_s^2/(4\pi)$:

$$\langle|\mathcal{M}|^2\rangle = \frac{4}{9}\,(4\pi\alpha_s)^2\,\frac{s^2+u^2}{t^2} \tag{30}$$

8 Cross-Check: Ratio of QCD to QED

$$\frac{\langle|\mathcal{M}|^2\rangle_{\text{QCD}}}{\langle|\mathcal{M}|^2\rangle_{\text{QED}}} = \frac{\frac{4}{9}g_s^4}{2\,g_e^4} = \frac{2}{9}\,\frac{g_s^4}{g_e^4} \tag{31}$$

Interpreting the Ratio

The factor $\frac{2}{9}$ is the **color-averaged color factor** $\langle|C|^2\rangle$. It encodes the "dilution" of the interaction due to averaging over 9 initial color states and summing over 9 final color states,

with only 8 gluons mediating between 3 colors.
The factor $\frac{g_s^4}{g_e^4}$ is the straight coupling constant replacement: strong force instead of electromagnetic. At typical parton-scattering energies ($Q \sim 100$ GeV), $\alpha_s \sim 0.1$ vs. $\alpha \sim 1/137$, so $g_s^4/g_e^4 \sim (0.1 \times 137)^2 \approx 190$. QCD scattering is much stronger!
The $\frac{s^2+u^2}{t^2}$ factor is unchanged because it comes entirely from the spin/momentum structure of two spin-$\frac{1}{2}$ particles exchanging a massless spin-1 boson. This kinematics is universal.

9 Summary: The Complete Map

Everything in One Place

	QED: $e^-\mu^- \to e^-\mu^-$	**QCD**: $q\,q' \to q\,q'$
Amplitude	$\frac{g_e^2}{t}[\bar{u}\gamma^\mu u][\bar{u}\gamma_\mu u]$	$\frac{g_s^2}{4t}[\bar{u}\gamma^\mu u][\bar{u}\gamma_\mu u]\sum_a \lambda^a_{ji}\lambda^a_{lk}$
Spin avg. factor	$\frac{1}{4}$	$\frac{1}{4}$
Color avg. factor	1 (no color)	$\frac{1}{9}$
Dirac traces	$8(s^2+u^2)$	$8(s^2+u^2)$ (identical)
Color sum	1	$\sum_{a,b}[\mathrm{Tr}(\lambda^a\lambda^b)]^2 = 32$
$\langle\lvert\mathcal{M}\rvert^2\rangle$	$\frac{2\,g_e^4(s^2+u^2)}{t^2}$	$\frac{4}{9}\,\frac{g_s^4(s^2+u^2)}{t^2}$